

UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The study of Higgs properties in the four-lepton final state at CMS

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Chapter 1

Probing four lepton final-state for Higgs Cross section measurement

1.1 Introduction

The ATLAS and CMS collaborations of CERN announced the discovery of much awaited Higgs boson in early July 2012 [1, 2] verifying the underlying concept of electroweak symmetry breaking (EWSB) in the SM of particle physics. This triggered essential precise studies of this particle to confirm its nature and to discover some new physics in case of some deviation while studying its properties. Later on, these experiments have also studied its comprehensive properties [3, 4, 5, 6] which are found to be consistent with the SM Higgs. There is a legacy of measurements on this topic from CMS and ATLAS collaborations in different LHC data taking eras for different decay channels of Higgs which can be studied in Ref. [Khachatryan:2015yvw, CMS:2015hja, CMS:2016rqf, Sirunyan:2017exp, CMS:2018hhg, atlas-conf-2019-029, atlas-conf-2019-025, atlas-conf-2019-032, CMS:2019drq, 7, 8, 9, 10, 11, 12, 13]. However, these measurements can be improved with newer data at higher luminosities and improved statistical techniques.

The cross section measurements (integrated and differential) are performed using collisions data of CMS (LHC) corresponding to 137.1 fb^{-1} integrated luminosity at $\sqrt{s} = 13 \text{ TeV}$. The studied differential observables are sensitive to production mechanism of Higgs. Further details follow in next sections. To minimize the model dependence, results are extracted in the fiducial volume that closely matches with the detector geometry as shown in Figure 1.1 [ref_fiducial]. Contribution of other model dependence is extracted by repeating the measurements for SM and other exotic models. The differences are reported as systematic uncertainty. All measurements includes the corrections for the finite detection efficiency

and resolution effects, and are extracted assuming $m_H = 125.09$ GeV, as measured from $H \rightarrow 2\gamma$ and $H \rightarrow 4\ell$ decay modes at CMS experiment of LHC [14].

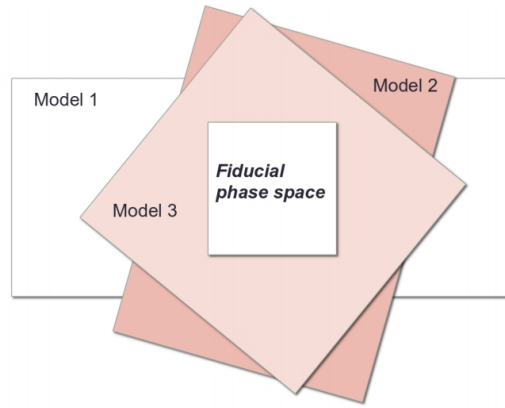


FIGURE 1.1: Fiducial phase definition among different models.

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1.2 Datasets and Triggers

The analysis uses full Run 2 (2016, 2017 and 2018) CMS data corresponding to 137.1 fb^{-1} integrated luminosity.

Used datasets are listed and documented in [CMS:2019chr] with the corresponding integrated luminosities. The analysis relies on four different primary datasets (PDs), *DoubleMuon*, *MuEG*, *EGamma*, and *SingleMuon*, each of which combines a certain collections of HLT paths. Events are collected to avoid duplications from different PDs, using:

- EGamma if they pass the diEle or triEle or singleElectron triggers,
- DoubleMuon if they pass the diMuon or triMuon triggers and fail the diEle and triEle triggers,
- MuEG if they pass the MuEle or MuDiEle or DiMuEle triggers and fail the diEle, triEle, singleElectron, diMuon and triMuon triggers,
- SingleMuon if they pass the singleMuon trigger and fail all the above triggers.

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The HLT paths used for full Run 2 collision data is listed in [CMS:2019chr] and also in the Appendix.

1.2.1 Trigger Efficiency

Efficiency in data of combination of triggers used for the analysis is measured by considering 4ℓ events triggered by single lepton triggers. Among the reconstructed four leptons, is geometrically associated with an object that passes at least one single electron or muon triggers. Rest of three leptons are regarded as “probes”. In a 4ℓ event, the possible tag-probe pairs are 4 and are counted in denominator while computing efficiency. For each of three leptons which are probes, all matching trigger filter objects are collected. Then the matched trigger filter objects of the three probe leptons are combined in attempt to reconstruct any of the triggers used in the analysis. If any of the analysis triggers can be formed using the probe leptons, probes are called passing probes which are counted as numerator for the efficiency measurement.

This technique is used to measure efficiency on MC and the comparison with data is exploited to estimate the reliability of simulation. Fig. 1.2 shows the measured efficiency both in data and MC (2018) varied with minimum p_T of leptons. MC efficiency seems to describe data well within the uncertainties.

1.3 Simulation Samples

Simulations of the interesting events help us build a realistic model of experiment in high energy physics. It enables us for a detailed study of physics processes (signal and background). In high energy physics experiments, different MC event generators are used to simulate different process. For this analysis, in particular, descriptions of the production of SM Higgs through process of gluon-gluon fusion (Fig. 1.3) are acquired using POWHEG V2 [powheg, powhegvv] generator. At accuracy level of next-to-next-to-leading order (i.e. NNLO) in perturbative QCD (pQCD), $gg \rightarrow H$ process description is obtained by partonic level MC generator NNLOPS.

The POWHEG, provides perturbative QCD calculations at next-to-leading (NLO) accuracy with a connection to parton-shower programs. Using the POWHEG, Higgs production via $gg \rightarrow H$ with zero associated jets can be described at accuracy level

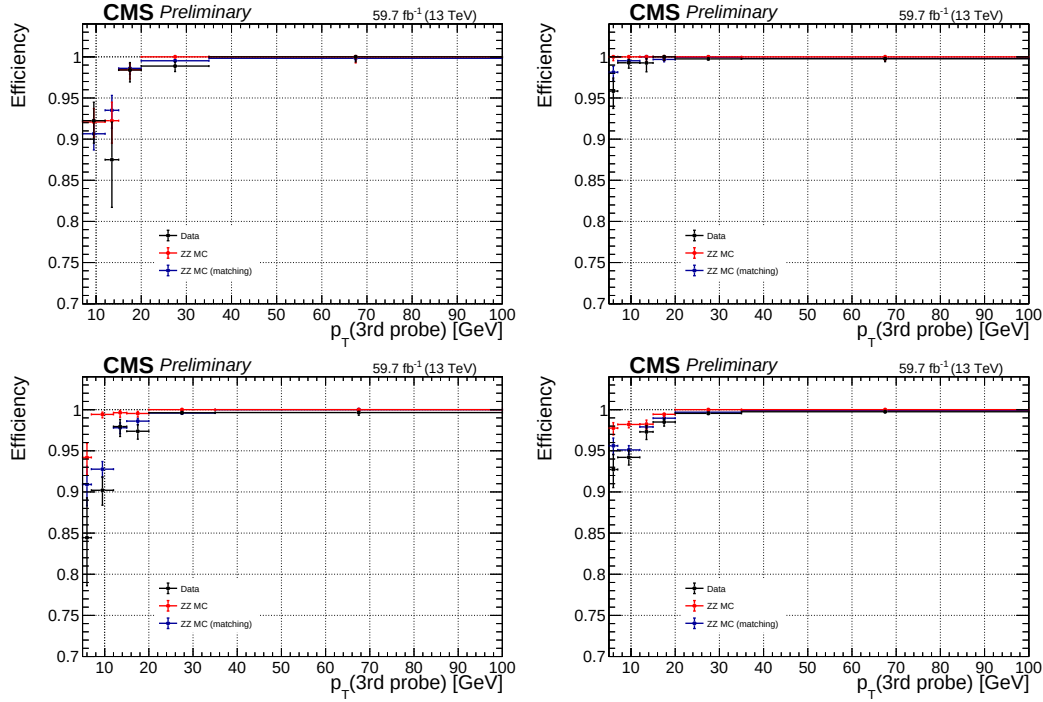


FIGURE 1.2: Measured trigger efficiency in 2018 data where the event collection is made using single electron and muon triggers: $4e$ (top left), 4μ (top right), $2e2\mu$ (bottom left) and 4ℓ (bottom right) final states.

of NLO. Finite masses of top and bottom quarks has been considered for $gg \rightarrow H$ production for all generators.

Signal Samples

Descriptions of the SM Higgs boson production are obtained using the POWHEG V2 [Alioli:2008gx, Nason:2004rx, Frixione:2007vw] generator for the five main production modes (ggH , VBF, WH, ZH and ttH) as described in section ?? [Bagnaschi:2011tu], [Nason:2009ai], [Hartanto:2015uka], [Luisoni:2013kna]. However, WH and ZH production modes are simulated by the MINLO HVJ extension of the POWHEG [Luisoni:2013kna]. Higgs decay to four lepton is simulated by JHUGEN generator [Gao:2010qx]. For the WH, ZH and ttH modes, Higgs is allowed to decay to $H \rightarrow ZZ \rightarrow 2\ell 2X$ such that two leptons from 4-lepton events are produced from the decay of associated W, Z bosons or top quarks respectively. Showering of parton-level events is done using PYTHIA8.209, and in all cases matching is performed by allowing QCD emissions at all energies in the shower and vetoing them afterwards

according to the POWHEG internal scale. All samples are generated with the default NNPDF 3.1 NLO parton distribution functions (PDFs) [Ball:2014uwa]. The list of signal samples with the corresponding cross sections are shown in [CMS:2019chr].

These samples, for each production mode, are also used to assess model dependence of measured results as shown in the Table ?? . The details of the method used to assess the model dependence will be described in Section ?? . The POWHEG and NNLOPS based samples for $gg \rightarrow H$ production are utilized as theoretical predictions based on SM calculations for the comparison with measured results as described in in Section ?? .

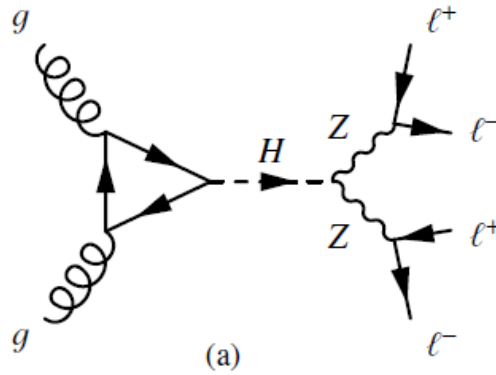


FIGURE 1.3: $gg \rightarrow H$ process feynman diagram.

All these MC samples are produced using POWHEG for the representation of QCD effects at NLO in Higgs production, and by using JHUGEN [Gao:2010qx, Bolognesi:2012mm, Anderson:2013afp] to represent such exotic resonances decays to four-leptons containing all the spin correlations.

Background Samples

For the estimations of the backgrounds, POWHEG V2 interfaced with PYTHIA8 was used to generate the dominant process $q\bar{q} \rightarrow ZZ \rightarrow 4\ell$ at the NLO accuracy level containing s-, t- and u-channel diagrams [Nason:2013ydw]. Generators were used with same configurations as used for the Higgs signal . Feynman diagrams of the involved background processes are shown in the Figure 1.4. The dynamical renormalization and factorization scales have been chosen equal to m_{ZZ} since the simulation covers a wide range of invariant masses of ZZ.

The $gg \rightarrow ZZ$ process, as shown in the Figure 1.4(c) is simulated at leading order (LO) with MCFM [MCFM, Campbell:2013una]. In order to have matching

in spectra of $gg \rightarrow H \rightarrow ZZ$ as predicted at NLO by POWHEG, the showering of these samples has been performed with some different PYTHIA8 settings. This only allows emissions only up to parton-level scale (“wimpy” shower).

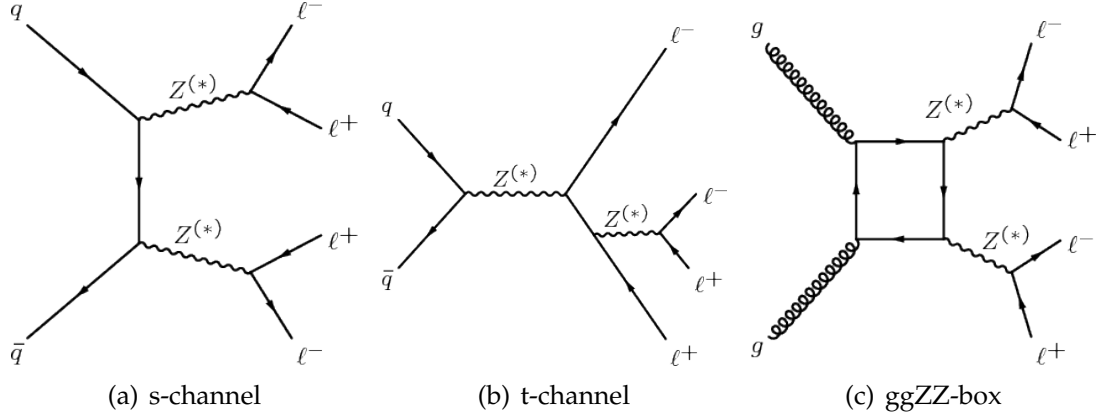


FIGURE 1.4: $q\bar{q} \rightarrow ZZ \rightarrow 4\ell$ process s- and t- channel fynman diagrams (a) and (b). $gg \rightarrow ZZ \rightarrow 4\ell$ process fynman diagram (c).

The initial-state and final-state radiations are included for all events generators, which provides hints for the existence of extra (hard) photons in the event. Interference between all amplitudes of process $H \rightarrow 4\ell$ are properly taken into consideration both in POWHEG and the JHUGEN. For the generators, the used default parton distribution functions (PDFs) are NNPDF3.1 NLO (at NLO and LO) are used.

The generated events were interfaced to PYTHIA 8.209 Tune tune CUETP8M1 to include the fact of the parton shower, underlying events and hadronization.

Processing of all generated events was made via a full simulation for CMS detector using GEANT4 [GEANT] and their reconstruction was done by exploiting same algorithms used for the data analysis. Multiple proton-proton interactions are incorporated in the simulations by reweighting the pileup distribution to match the distribution of the observed data.

Pileup Reweighting

For each year, corresponding simulation samples are reweighted to meet pileup distribution in the data. An example of the reweighting procedure for 2018 data is shown in Fig. 1.5.

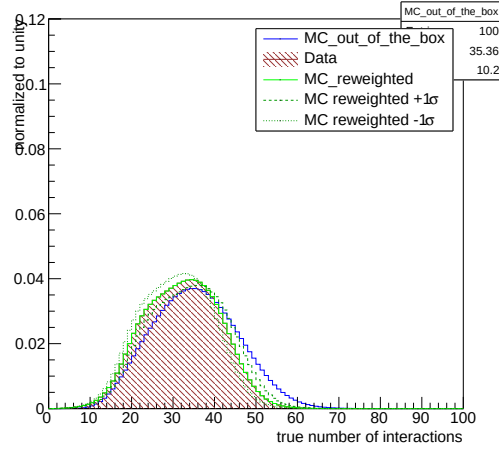


FIGURE 1.5: Pileup distribution in 2018 MC and Data shown before and after the application of PU weights. Up and down variations of 5% in the minimum bias cross section when calculating the weights are also shown.

1.4 Analysis Strategy

As mentioned in Section 1.3, the measurements on Higgs cross sections (Inclusive and differential) are made in a fiducial region that matches the reconstruction level selection. Measurement strategy and definition of the fiducial volume is optimized to:

- detect a low mass Higgs (m_H 125.09 GeV) boson decaying to 4 leptons through a pair of Z bosons
- keep the measurements independent of how the Higgs boson is produced.

For reconstruction efficiency to be independent of fiducial phase space, it is necessary to include isolation in the definition of fiducial phase space.

In case of associated production of Higgs with vector bosons where the bosons can decay leptonically, there is possibility that Higgs is reconstructed from wrong combination of leptons. This makes it necessary to consider such “wrong combination” events as a background in order to keep complete information from unbinned simultaneous fit of signal and background parameterizations to four-lepton invariant mass.

Also, signal events which pass the fiducial level selections but fail the reconstruction are accounted in the fiducial efficiency (ϵ) and the events which pass the

reconstruction but fail the fiducial selections are referred to as “nonfiducial signal” events (f_{nonfid}) and are treated as a background as illustrated in Figure 1.6 [fid_diagram].

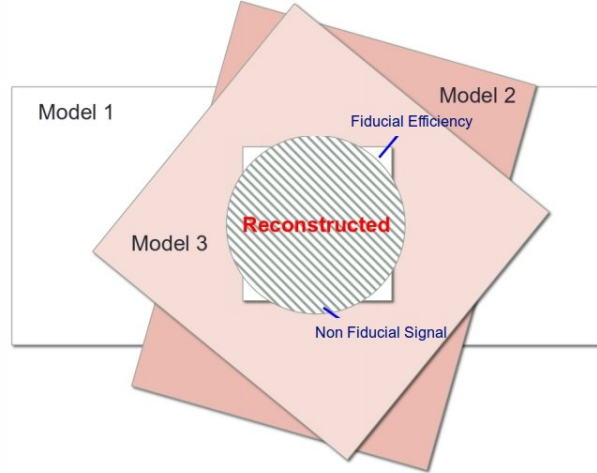


FIGURE 1.6: Fiducial efficiency(ϵ) and nonfiducial(f_{nonfid}) signal definition.

There is significant effect of imperfect resolution of the detector on the differential measurements particularly in case of jet related observables for which detector resolution is poor. A procedure, called unfolding, is applied for the correction of the reconstruction efficiencies and the effects are applied to the measurements.

The effects of imperfect detector resolution can in general have a non-negligible effect on shape of the distributions of measured observables. For this reason, for differential cross section measurements of Higgs, a procedure to correct the detection efficiencies and resolution effects is applied.

The details of the procedure would be described in next Sections.

1.4.1 Event Selection

This dissertation is based on the analysis for which event selections are similar to what was used in pervious analyses on same topic using the same final state [CMS:2018mmw, CMS:2017jkd, Chatrchyan:2013mx, Khachatryan:2014kca]. The online event selection is based on having fired High Level Trigger paths as described in 1.2. Unlike in the analysis based on Run I dataset, we require OR of

all HLT paths so that the trigger requirement does not depend on the selected final state. The reason is Full Run II data (at 137.1 fb^{-1}), associated production modes that can come with additional leptons are targeted which improve trigger efficiency.

Vertex Selection

The events are required to have at least one primary vertex (PV) that fullfills the criteria:

- Large degree of freedom ($N_{PV} > 4$)
- Small radius of PV being small ($r_{PV} < 2 \text{ cm}$)
- Restricted collisions along z -axis ($z_{PV} < 24 \text{ cm}$).

p_T threshold requirements on the Triggers

For the double lepton triggers, the requirement on minimum p_T for leading and the subleading lepton are imposed to be 17 and 8 GeV respectively; however, for triple electron triggers, it is 15, 8 and 5 GeV in the same order respectively.

Lepton Selection

Details on the selection requirments on electron and muons are explained in sections ?? and ??, and ?? and ?? respectively. Both electrons and muons are initially required to pass POG Loose identification criteria with subsequent requirements on imparameter and isolation. These leptons are regarded as **selected leptons** and can be used for analysis.

Jet Selection

Reconstruction of the jets is mandatory to study the differential cross section measurements for the jet related observables. Jets are reconstructed with $p_T \geq 30 \text{ GeV}$ within $|\eta| \leq 4.7$, discussed in Section ?? . To remove overlapping of jets overlap from leptons or selected photons, it is required to have at least $\Delta R > 0.5$ where $\Delta R \equiv \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$ [Chatrchyan:2013mx] with respect to any of the selected leptons and FSR photons candidates.

1.4.2 ZZ Candidate Selection

Four-lepton candidates are built from the selected leptons, that pass the $\text{SIP}_{3\text{D}} < 4$ vertex constraint and the isolation cuts (defined in sections ?? and ??), where FSR photons are subtracted as briefed in Section ?. Cross cleaning of the leptons is made by discarding electrons which are within $\Delta R < 0.05$ of selected muons.

The construction and selection of four-lepton candidates proceeds according to the following sequence:

1. **Z candidates** are built from oppositely charged and same flavoured leptons (e^+e^- , $\mu^+\mu^-$) with additional requirement on dilepton mass as $12 < m_{\ell\ell(\gamma)} < 120\text{GeV}$. Z boson candidate is also required to include FSR photons (if any).
2. **ZZ candidates** are then built from non-overlapping Z boson candidates. The one who is reconstructed with $m_{\ell\ell}$ closer to the SM Z boson mass is regarded as Z_1 and other as Z_2 . Additional requierements on ZZ candidates are as under:
 - **Ghost removal** : Selected four leptons are required to have $\Delta R(\eta, \phi) > 0.02$ among each other.
 - **lepton p_T** : Among selected four leptons, two are required to fulfill $p_{T,i} > 20\text{GeV}$ and $p_{T,j} > 10\text{GeV}$.
 - **QCD suppression**: Dilepton inivariant mass is required to pass (regardless of flavor) $m_{\ell\ell} > 4\text{GeV}$ for QCD induced J/Ψ veto.
 - **Z_1 mass**: $40\text{GeV} < m_{Z1} < 120\text{GeV}$
 - **four-lepton invariant mass**: $m_{4\ell} > 70\text{GeV}$
3. **Signal region** is built from the events containing one selected ZZ candidate at leaset.

1.4.3 Choice of best ZZ Candidate

An important change with respect to the Run I analysis is to choose the best ZZ candidate after all kinematic cuts. This helps to test other selection strategies for this candidate choice and becomes important for events having more than four selected leptons to search for Higgs boson production modes where associated particles can decay leptonically (e.g. VH and ttH).

For fiducial cross section measurement, Z_1 is chosen with mass closest to the mass of SM Z boson mass while Z_2 is chosen from Z candidates whose lepton pair gives higher p_T sum.

1.4.4 Definition of the Fiducial Volume

The measurements on Higgs cross sections (Inclusive and differential) are made in a fiducial region that matches the reconstruction level selection. This is done because fraction of all $H \rightarrow 4\ell$ events which pass the selection criteria as described in Section 1.4.3 can significantly differ for different Higgs production mechanisms and also for its distinct exotic models. As a treatment for the impact of model dependent acceptances, these measurements are done in the a certain fiducial volume which is defined so as to closely match with the experimental acceptance from reconstruction level selections.

Requirement of fiducial phase space is minimum four leptons having $p_T > 7$ GeV and $|\eta| < 2.5$ in case of electron, and $p_T > 5$ GeV and $|\eta| < 2.4$ in case of muon. Besides, the leading and sub-leading leptons are required to have $p_T > 20$ GeV and $p_T > 10$ GeV respectively. The four-momenta of each lepton is calculated at the hard scattering level (PYTHIA status code = 3), before any final state radiation occurs. Each lepton isolation is calculated by adding p_T of all stable particles (PYTHIA status code = 1) within a cone size $\Delta R < 0.3$ around the lepton. All neutrinos and FSR photons are excluded from above summation. Then the lepton's relative isolation which is defined by ratio of this sum and p_T of considered lepton and is required to be less than 0.35, consistent with reconstruction level isolation requirement [Chatrchyan:2013mxa]. The studies based on simulation samples show that signal efficiency can significantly change among different models without requiring lepton isolation at fiducial level as shown in Table ??, the maximum difference comes from $t\bar{t}H$ and $gg \rightarrow H$ production modes. This behavior is expected as $t\bar{t}H$ events has more hadronic activity around leptons as compared to $gg \rightarrow H$ events, so the cut at relative isolation kills more events in $t\bar{t}H$ than $gg \rightarrow H$. Hence, for reconstruction efficiency to be independent of fiducial phase space, it is necessary to include isolation in the definition of fiducial phase space.

Furthermore, fiducial phase space is defined by several other topological and kinematics selections. There must be two SFOS lepton pairs where Z_1 being the one with invariant mass closest to the SM Z boson and reconstructed with $40 \leq$

$m(Z_1) \leq 120$ GeV. Z_2 is reconstructed by remaining SFOS pair of leptons within inv.
 mass range of $12 \leq m(Z_2) \leq 120$ GeV. In case of more than one Z_2 passing the crite-
 ria, leptons pair with highest $\Sigma|p_T|$ is selected. To remove the overlapping between
 leptons, we ask for $\Delta R(\ell_i \ell_j) > 0.02$ for any $i \neq j$. To remove contamination from
 QCD low masses resonance, any opposite-charge pair of leptons is required to
 meet the condition $m(\ell^+ \ell'^-) \geq 4$ GeV. Finally, We require that all selected leptons
 must emerge from Higgs boson decay and invariant mass of four-leptons must ful-
 fill $105 \text{ GeV} < m(4\ell) < 140 \text{ GeV}$. Demand for $m(4\ell)$ helps to control the contam-
 ination from off-shell gg fusion production which is of considerable size, capable
 to increase the total cross section by a few percent [Kauer:2013cga, Kauer:2012hd].
 The requirements that defines fiducial phase space are also gathered in Table 1.1.

TABLE 1.1: Overview of all requirements utilized to define the fiducial phase space.

Requirements on $H \rightarrow 4\ell$ fiducial phase space	
Lepton kinematics and the isolation	
p_T of the leading lepton	$p_T > 20 \text{ GeV}$
p_T of sub-leading lepton	$p_T > 10 \text{ GeV}$
Additional electrons (muons) p_T	$p_T > 7$ (5) GeV
Pseudorapidity of electrons (muons)	$ \eta < 2.5$ (2.4)
Sum of the scalar p_T of all the stable objects within $\Delta R < 0.3$ from the lepton	$< 0.35 p_T$
Event topology	
From leptons satisfying criteria above, minimum two SFOS lepton pairs	
Invariant mass of Z_1 candidate	$40 < m(Z_1) < 120 \text{ GeV}$
Invariant mass of Z_2 candidate	$12 < m(Z_2) < 120 \text{ GeV}$
Separation between selected four leptons	$\Delta R(\ell_i \ell_j) > 0.02$
Invariant mass of any OS lepton pair	$m(\ell_i^+ \ell_j^-) > 4 \text{ GeV}$
Invariant mass of selected four leptons	$105 < m_{4\ell} < 140 \text{ GeV}$

It has been established based on simulation samples that with including isola-
 tion for the events passing fiducial phase space definition reconstruction efficien-
 cies faintly depends on Higgs properties and its production modes. The remaining
 model dependence will be quoted as a separate uncertainty as explained in Section
 ?? . Part of the signal events passing fiducial phase space definition ($\mathcal{A}_{\text{fid}}$), and re-
 construction efficiencies (ϵ) for SM production modes are registered in Table 1.2.

TABLE 1.2: Summary of different SM signal models. The MC samples are from full Run 2 yearly productions (2016, 2017 and 2018) assuming $m_H = 125.0$ GeV. Luminosity averaged fiducial acceptance ($\mathcal{A}_{\text{fid}}$) defined the fraction of events passing fiducial phase space selections, reconstruction efficiency (ϵ) represented by the ratio of reconstructed to accepted events. Also, f_{nonfid} which are reconstructed events but does not pass fiducial phase space definition at generator level. Values are obtained using 13 TeV signal models and the uncertainties shown are only statistical uncertainties.

Signal process	$\mathcal{A}_{\text{fid}}$	ϵ	f_{nonfid}	$(1 + f_{\text{nonfid}})\epsilon$
Individual production modes of Higgs boson				
gg→H (POWHEG)	0.402 ± 0.001	0.592 ± 0.002	0.053 ± 0.001	0.624 ± 0.002
VBF (POWHEG)	0.444 ± 0.002	0.605 ± 0.003	0.043 ± 0.001	0.631 ± 0.003
WH (POWHEG+MINLO)	0.325 ± 0.002	0.588 ± 0.003	0.075 ± 0.002	0.632 ± 0.004
ZH (POWHEG+MINLO)	0.340 ± 0.003	0.594 ± 0.005	0.081 ± 0.004	0.643 ± 0.006
ttH (POWHEG)	0.314 ± 0.003	0.585 ± 0.006	0.169 ± 0.006	0.684 ± 0.007

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